Linearized Drift-Pressure Model: Definitions, Equilibrium, and the Linear Stability Matrix

Contents

Overview

This note collects the working derivation into a single, self-contained reference. Section 1 fixes notation and lists every constant/operator that appears, with its physical meaning and (where defined) its formula. Section 2 states the two governing equations. Section 3 derives the axisymmetric, steady equilibrium. Section 4 introduces the perturbation ansatz and linearizes both equations term by term. Section 5 assembles the 2×2 linear operator (eigenvalue) problem, giving *both* the compact and the fully expanded operator entries. Section 6 collects every approximation and open question.

1 Notation, Operators, and Constants

1.1 Bracket and Laplacian operators

Generalized Poisson bracket. For two scalar fields $a(r,\theta)$, $b(r,\theta)$,

$$\{a,\, b\} \;\equiv\; a_x b_y - a_y b_x, \tag{1}$$

where subscripts denote partial derivatives ($a_x = \partial a/\partial x$, etc.). In polar coordinates this is equivalently

$$\{a,\, b\} = \frac{1}{r}(a_r b_\theta - a_\theta b_r)\,. \tag{2}$$

Laplacian.

$$\Delta \;\equiv\; \nabla^2. \tag{3}$$

When axisymmetry is assumed (no θ-dependence in the background), the full Laplacian acting on a perturbation $\propto e^{im\theta+\gamma t}$ separates as

$$\Delta \;\to\; \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}, \tag{4}$$

and we define the *m-th harmonic radial Laplacian operator*

$$\Delta_m \;\equiv\; \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) - \frac{m^2}{r^2}, \tag{5}$$

i.e. $\Delta\big(\hat\varphi\, e^{im\theta+\gamma t}\big) = \Delta_m \hat\varphi\, e^{im\theta+\gamma t}$, since $\partial_\theta^2 \to -m^2$ on this harmonic.

Note: for an equilibrium quantity $f_0(r)$ that has *no* θ-dependence, the relevant Laplacian is the $m = 0$ case, $\Delta f_0 = \frac{1}{r}\frac{d}{dr}\Big(r\frac{df_0}{dr}\Big) = f_0'' + \frac{f_0'}{r}$, *not* $\Delta_m f_0$ (which would add a spurious $-m^2/r^2$ term).

1.2 Field variables

Symbol	Meaning
φ	Normalized plasma potential, $\varphi = \phi/\phi_\star$, where $\phi_\star$ is the (central) amplitude of the applied wall potential.
P	Normalized pressure, $P = p/p_0$, where p_0 is the plasma pressure on axis at the trap midplane.
ρ	Mass density, $\rho = m_i n_0$ (ion mass $\times$ equilibrium number density).

1.3 Constants and coefficients

Symbol	Definition	Physical meaning
U	$U = \dfrac{p_{0i}}{e\,n\,\phi}$	FLR parameter (ion pressure / charge density × potentia
H	$H = \dfrac{\tau^2 e\phi}{cLT_e}\left(\dfrac{J_{i0}}{B}\right)_{\text{wall}}\dfrac{1}{\langle \rho B^{-2}\rangle}$	Plasma–wall coupling (line-tying) strength.
$\mathcal{K}$	$\mathcal{K} = 2P_0\tau^2\,\dfrac{\langle g(z)\,\alpha(z)\,B^{-1}\rangle}{\langle \rho B^{-2}\rangle}$	Normalized average magnetic curvature.
J_{i0}	—	Equilibrium axial ion current.
τ	$\tau = \dfrac{BR^2}{c\phi}$	$E\times B$ drift period (BR^2 is const. on each field line).
L	—	Half-length of the trap.
$g(z)$	$\langle g\rangle = 1$	Axial pressure profile.
$\alpha(z)$	$\dfrac{1}{B(r,z)} = \dfrac{1}{B(0,z)} + \alpha(z)\,r^2$	Paraxial field expansion coefficient; obtainable from $\nabla\cdot\mathbf{B}$
T_e	—	Electron temperature.
φ_{fe}	—	Floating potential (flat in radius at constant T_e).
φ_{wall}	—	End-wall bias (applied externally).
φ_w	$\varphi_w = \varphi_{fe} + \varphi_{\text{wall}}$	Total effective wall potential.
$\nu_\perp^P$	—	Collisional transverse (cross-field) diffusion rate acting on
$\nu_\parallel^P$	—	Axial diffusion rate acting on P.
$\nu_\perp$, $\nu_\parallel$	—	Transverse/axial diffusion rates acting on φ (not separate
Q_P	$Q_P \approx Q_P(r)$	Power source term in the pressure equation.
S_p	$S_p \approx (\nu_\perp^P - \nu_\perp)\Delta P - (\nu_\parallel^P - \nu_\parallel)P + Q_p$	Effects of external heating, axial outflow, and transverse t

Open notational issue: it is not fully specified whether φ_w is normalized with respect to ϕ in the same way as φ, i.e. whether $\varphi_w \equiv \phi_w/\phi_\star$ consistently.

2 Governing Equations

The starting point is a pair of coupled evolution equations for the normalized potential φ and normalized pressure P:

$$\partial_t\Delta\varphi + \{\varphi, \Delta\varphi\} - U\nabla\cdot\{\nabla\varphi, P\} = H(\varphi - \varphi_w) + \mathcal{K}\{P, r^2\} + \nu_\perp\Delta^2\varphi - \nu_\parallel\Delta\varphi - U\Delta S_P, \quad (6)$$

$$\partial_t P + \{\varphi, P\} = \nu_\perp^P\Delta P - \nu_\parallel^P P + Q_P. \quad (7)$$

Equation (6) is the vorticity (potential) equation; Equation (7) is the pressure equation.

3 Equilibrium

Assumption E1 (axisymmetric, steady equilibrium). The background state has no θ-dependence and is time-independent: $\partial_t(\cdot)_0 = 0$ and $\partial_\theta(\cdot)_0 = 0$.

Under E1, any bracket of two equilibrium quantities vanishes since both fields are functions of r only, so every θ-derivative is zero. Equation (6) therefore collapses to

$$0 = H(\varphi_0 - \varphi_w) + \nu_\perp \Delta^2 \varphi_0 - \nu_\parallel \Delta \varphi_0 - U \Delta S_{P,0}. \tag{8}$$

Assumption E2. The diffusive terms $\nu_\perp \Delta^2 \varphi_0$ and $\nu_\parallel \Delta \varphi_0$ are small compared to $H(\varphi_0 - \varphi_w)$.

Assumption E3. The axial and transverse diffusion rates for φ and P coincide: $\nu_\perp \approx \nu_\perp^P$ and $\nu_\parallel \approx \nu_\parallel^P$.

With E2 and E3 the equilibrium relation reduces to

$$H(\varphi_0 - \varphi_w) + U \Delta Q_p = 0 \quad \Longrightarrow \quad \boxed{\varphi_0 \approx \varphi_w.} \tag{9}$$

The equilibrium condition for the *pressure* equation (Eq. (7) with E1 and Assumption L3 below) is

$$\nu_\parallel^P P_0 = Q_p \qquad \text{(equilibrium pressure balance)}. \tag{10}$$

3.1 Equilibrium rotation frequency

Since $\varphi_0 = \varphi_0(r)$ the equilibrium $E \times B$ flow is purely azimuthal with frequency

$$\Omega_0(r) \;\equiv\; \frac{1}{r}\frac{\partial \varphi_0}{\partial r} \;\propto\; \frac{1}{r}\frac{\partial \varphi_w}{\partial r}, \tag{11}$$

where the second equality uses Eq. (9).

4 Perturbation Ansatz and Linearization

Assumption L1 (normal-mode ansatz).

$$\varphi(r,\theta,t) = \varphi_0(r) + \hat{\varphi}(r)\, e^{im\theta + \gamma t}, \tag{12}$$

$$P(r,\theta,t) = P_0(r) + \hat{p}(r)\, e^{im\theta + \gamma t}. \tag{13}$$

Here $m \in \mathbb{Z}$ is the azimuthal mode number and $\gamma \in \mathbb{C}$ is the growth rate.

Assumption L2 (linearization). Only first-order terms in the perturbation amplitudes $(\hat{\varphi}, \hat{p})$ are retained; products of two perturbed quantities are dropped.

4.1 Linearized pressure equation

Starting from Eq. (7) and applying L1–L2 with Assumption L3 below.

Assumption L3. The cross-field diffusion term $\nu_\perp^P \Delta P$ is dropped in the linearized pressure balance.

The bracket splits into:

$$\{\varphi_0, \delta P\} = \frac{1}{r}\frac{\partial \varphi_0}{\partial r}\frac{\partial(\delta P)}{\partial \theta} = im\,\Omega_0(r)\,\hat{p}\,e^{im\theta+\gamma t}, \tag{14}$$

$$\{\delta\varphi, P_0\} = -\frac{1}{r}\frac{\partial(\delta\varphi)}{\partial\theta}\frac{\partial P_0}{\partial r} = -\frac{im}{r}\,P_0'(r)\,\hat{\varphi}\,e^{im\theta+\gamma t}. \tag{15}$$

After separating orders in $e^{im\theta+\gamma t}$:

Order-zero (equilibrium): Eq. (10) is recovered: $\nu_\parallel^P P_0 = Q_p$.

Order-one (perturbed):

$$\gamma\hat{p} + im\Omega_0\,\hat{p} - \frac{im}{r}P_0'(r)\,\hat{\varphi} = -\nu_\parallel^P\hat{p}, \tag{16}$$

which rearranges to

$$\boxed{\gamma\hat{p} = \frac{im}{r}P_0'(r)\,\hat{\varphi} - \big(\nu_\parallel^P + im\Omega_0\big)\hat{p}.} \tag{17}$$

4.2 Linearized vorticity equation

Now linearize Eq. (6).

Assumption L4. The hyper-diffusive term $\nu_\perp \Delta^2\varphi$ and the source-gradient term $U\Delta S_P$ are dropped in the linear stability analysis.

Assumption L5. Products of two perturbed quantities ("frightening" terms) are discarded (part of strict linearization L2).

Term 1: $\partial_t \Delta\varphi$

$$\partial_t \Delta\varphi \;\longrightarrow\; \gamma\Delta_m\hat{\varphi}\,e^{im\theta+\gamma t}. \tag{18}$$

Term 2: $\{\varphi, \Delta\varphi\}$

Expand $\Delta\varphi = \Delta\varphi_0 + \Delta_m \delta\varphi$. The bracket splits into three pieces; the equilibrium–equilibrium piece vanishes by E1:

$$\{\varphi_0, \Delta\varphi_0\} = 0, \tag{19}$$

$$\{\delta\varphi, \Delta\varphi_0\} = -\frac{im}{r}\hat{\varphi}\, e^{im\theta+\gamma t}\,(\Delta\varphi_0)', \tag{20}$$

$$\{\varphi_0, \Delta\delta\varphi\} = im\,\Omega_0\,\Delta_m \hat{\varphi}\, e^{im\theta+\gamma t}. \tag{21}$$

Here primes denote d/dr, and $(\Delta\varphi_0)'$ is the radial derivative of $\Delta\varphi_0 = \varphi_0'' + \varphi_0'/r$ (the *full* Laplacian of the equilibrium potential, with no m^2/r^2 correction).

Term 3: $U\nabla \cdot \{\nabla\varphi, P\}$

Using $\nabla A = \hat{r}\,\partial_r A + \hat{\theta}\,(1/r)\partial_\theta A$ and $\nabla \cdot \mathbf{A} = (1/r)\partial_r(rA_r) + (1/r)\partial_\theta A_\theta$, the two linearized contributions are

$$\nabla \cdot \{\nabla\varphi_0, \delta P\} = \frac{1}{r}\frac{\partial}{\partial r}\big[im\,\hat{p}\,\varphi_0''\big]\, e^{im\theta+\gamma t}, \tag{22}$$

$$\nabla \cdot \{\nabla\delta\varphi, P_0\} = \frac{1}{r}\frac{\partial}{\partial r}\big[-im\,P_0'\,\hat{\varphi}'\big]\, e^{im\theta+\gamma t} + \frac{im^3}{r^3}P_0'\,\hat{\varphi}\, e^{im\theta+\gamma t}, \tag{23}$$

which combine to

$$\nabla \cdot \{\nabla\varphi, P\}\Big|_{\text{linear}} = \left[\frac{1}{r}\frac{\partial}{\partial r}\big(im\,\hat{p}\,\varphi_0'' - im\,P_0'\,\hat{\varphi}'\big) + \frac{im^3}{r^3}P_0'\,\hat{\varphi}\right] e^{im\theta+\gamma t}. \tag{24}$$

Term 4: $\mathcal{K}\{P, r^2\}$

Since r^2 has no θ-dependence, only $\{\delta P, r^2\}$ survives:

$$\mathcal{K}\{P, r^2\}\Big|_{\text{linear}} = -2im\mathcal{K}\,\hat{p}\, e^{im\theta+\gamma t}. \tag{25}$$

Assembled linearized vorticity equation

Collecting all terms, using $\varphi_0 \approx \varphi_w$ (so $H(\varphi_0 - \varphi_w) \to 0$ for the equilibrium and $H\hat{\varphi}$ for the perturbation), and dividing by $e^{im\theta+\gamma t}$:

$$\begin{aligned}\gamma\Delta_m\hat{\varphi} \;=\; & -im\Omega_0\,\Delta_m\hat{\varphi} + \frac{im}{r}(\Delta\varphi_0)'\,\hat{\varphi} - \frac{imU}{r}\frac{\partial}{\partial r}\left(P_0'\hat{\varphi}'\right) + \frac{im^3U}{r^3}P_0'\hat{\varphi} + H\hat{\varphi} - \nu_\parallel\,\Delta_m\hat{\varphi} \\ & + \frac{imU}{r}\frac{\partial}{\partial r}\left(\varphi_0''\hat{p}\right) - 2im\mathcal{K}\hat{p}.\end{aligned}$$

Here $(\Delta\varphi_0)' = \varphi_0''' + \varphi_0''/r - \varphi_0'/r^2$ is the radial derivative of the equilibrium Laplacian $\Delta\varphi_0 = \varphi_0'' + \varphi_0'/r$. This term represents the shear in equilibrium vorticity and is kept as an active contribution.

5 The Linear Eigenvalue System

Equations (17) and (4.2) are linear and homogeneous in $(\Delta_m\hat{\varphi},\, \hat{p})$. Writing the system as a and eigenvalue γ:

$$\gamma \begin{pmatrix} \Delta_m & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix} = \begin{pmatrix} \mathcal{L}_{11} & \mathcal{L}_{12} \\ \mathcal{L}_{21} & \mathcal{L}_{22} \end{pmatrix} \begin{pmatrix} \hat{\varphi} \\ \hat{p} \end{pmatrix}. \tag{26}$$

5.1 Compact (as-derived) operator entries

The entries read directly from Eqs. (17) and (**??**), using Δ_m^{-1} to convert any $\hat{\varphi}$ back to $\Delta_m\hat{\varphi}$:

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' \; - \frac{imU}{r}\frac{\partial}{\partial r}\big(P_0'\,\cdot\big) + \frac{im^3U}{r^3}\,P_0' \; + H \; - \nu_\parallel, \Delta_m, \tag{27}$$

$$\mathcal{L}_{12} = \frac{imU}{r}\frac{\partial}{\partial r}\big(\varphi_0''\,\cdot\big) - 2im\mathcal{K}, \tag{28}$$

$$\mathcal{L}_{21} = \frac{im}{r}P_0'(r) \tag{29}$$

$$\mathcal{L}_{22} = -\nu_\parallel^P - im\Omega_0. \tag{30}$$

5.2 Expanded operator entries (product rule applied explicitly)

$\mathcal{L}_{11}$ expanded

Apply the product rule to the FLR term:

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' \; - \frac{imU}{r}\frac{\partial}{\partial r}\big(P_0'\,\cdot\big) + \frac{im^3U}{r^3}\,P_0' \; + H \; - \nu_\parallel\Delta_m \tag{31}$$

$$\mathcal{L}_{11} = -im\Omega_0\Delta_m + \frac{im}{r}(\Delta\varphi_0)' - \frac{imU}{r}P_0'' + P_0'\,\frac{\partial}{\partial r} + \frac{im^3U}{r^3}\,P_0' + H - \nu_\parallel\Delta_m \tag{32}$$

where $(\Delta\varphi_0)' = \varphi_0''' + \varphi_0''/r - \varphi_0'/r^2$.

$\mathcal{L}_{12}$ expanded

$\mathcal{L}_{12}$ acts on $\hat{p}$:

$$\begin{aligned} \mathcal{L}_{12}\,\hat{p} &= \frac{imU}{r}\frac{d}{dr}\big(\varphi_0''\,\hat{p}\big) - 2im\mathcal{K}\,\hat{p} \\ &= \frac{imU}{r}\big(\varphi_0'''\,\hat{p} + \varphi_0''\,\hat{p}'\big) - 2im\mathcal{K}\,\hat{p}. \end{aligned} \tag{33}$$

$\mathcal{L}_{21}$ and $\mathcal{L}_{22}$

Both are already pure multiplication operators (no derivatives):

$$\mathcal{L}_{21}\,\hat{\varphi} = \frac{im}{r}P_0'(r)\cdot\hat{\varphi}, \tag{34}$$

$$\mathcal{L}_{22}\,\hat{p} = \big(-\nu_\parallel^P - im\Omega_0(r)\big)\cdot\hat{p}. \tag{35}$$

6 Summary of Approximations and Open Questions

6.1 Equilibrium approximations

- **(E1)** Axisymmetric, steady background: $\partial_\theta(\cdot)_0 = 0$, $\partial_t(\cdot)_0 = 0$. Kills every equilibrium–equilibrium bracket.
- **(E2)** Diffusion subdominant at equilibrium: $\nu_\perp \Delta^2 \varphi_0$ and $\nu_\| \Delta \varphi_0$ dropped relative to $H(\varphi_0 - \varphi_w)$, giving $\varphi_0 \approx \varphi_w$.
- **(E3)** $\nu_\| \approx \nu_\|^P$ and $\nu_\perp \approx \nu_\perp^P$ (equal diffusion rates for φ and P).
- *Open question:* Is $\nu_\| \approx \nu_\|^P$ valid?
- *Open question:* Is $\nu_\perp \ll \nu_\|$ (transverse diffusion negligible)?
- *Open question:* Is φ_w normalized consistently with φ (both divided by $\phi_\star$)?
- *Open question:* Can the axial pressure profile be approximated as $g(z) = 1$, given $\langle g \rangle = 1$ by definition?

6.2 Perturbation ansatz and linearization

- **(L1)** Single-harmonic normal-mode ansatz.
- **(L2)** Strict linearization: only terms first-order in $(\hat{\varphi}, \hat{p})$ kept.
- **(L3)** $\nu_\perp^P \Delta P \to 0$ in the linearized pressure balance.
- **(L4)** $\nu_\perp \Delta^2 \varphi$ and $U \Delta S_P$ dropped in the vorticity equation.
- **(L5)** No second-order ("frightening") terms.

6.3 Undefined or implicitly defined quantities

- $\nu_\perp$ and $\nu_\|$ (diffusion rates in the vorticity equation) are used but never explicitly defined in the source material.
- $\alpha(z)$ is defined implicitly via $1/B(r,z) = 1/B(0,z) + \alpha(z) r^2$ and must be computed from $\nabla \cdot \mathbf{B} = 0$ with the known on-axis field $B(0,z)$.